# TECHNICAL EVALUATION REPORT: LINEAR LAD MODEL

*Detailed Specification & Multi-Circuit Performance Benchmark for Emily's Model (Sprint 3 Task 6)*

**Model Owner:** Linear_LAD | **Prepared By:** Data Science Team (Task 6 Lead) | **Date:** November 2026

## 1. MODEL ARCHITECTURE & SPECIFICATIONS

The model uses a single pooled Least Absolute Deviation (LAD / median regression) algorithm regularized with $\alpha = 0.001$. Built without circuit identity, it is trained exclusively on the 8 NE/NW circuits and predicts across all 16 energy circuits.

- **Objective Function:** L1 Norm loss minimization (robust against isolated outlier load spikes).
- **Features:** 15 standardized features from `task8_model.FEATURES` including lagged consumption values and calendar embeddings.
- **Environment Specs:** Python 3.13, scikit-learn 1.9.1, pandas 3.0.5, numpy 2.5.3.

## 2. BENCHMARK RESULTS & COMPARATIVE PERFORMANCE

Results on the 8 shared NE/NW circuits (test MAE in Wh) across operational horizons:

| Horizon | Linear LAD Model | B3 Persistence | Random Forest | Gradient Boosting | Leading Model |
|---|---|---|---|---|---|
| +1h | **33.3 Wh** | 34.3 Wh | 35.9 Wh | 38.1 Wh | Linear Regression (Significantly Better) |
| +6h | 48.5 Wh | 82.9 Wh | **47.1 Wh** | 51.8 Wh | Random Forest (Statistically Tied) |
| Day-Ahead | 48.6 Wh | 57.4 Wh | **46.7 Wh** | 50.7 Wh | Random Forest (Statistically Tied) |

## 3. ZONE-BY-ZONE STATISTICAL BREAKDOWN (2,000 RESAMPLE BOOTSTRAP)

Statistical significance evaluation using 24-hour block bootstrap (2,000 resamples):

- **NE & NW Zones:** Linear Regression and Random Forest are statistically tied across all horizons.
- **SE Zone (+1h):** Linear Regression achieves a statistically significant victory (**30.5 Wh** vs 45.9 Wh for RF).
- **SE Zone (+6h & Day-Ahead):** Random Forest achieves a statistically significant victory (42.8 Wh vs 53.1 Wh at +6h; 42.6 Wh vs 55.5 Wh Day-Ahead).
- **SW Zone:** Statistically tied across all horizons.

## 4. VISUAL EVALUATION & ERROR BOUNDS




Figure 1: Comparative MAE metrics with 2,000-resample 24h block bootstrap 95% confidence intervals across horizons.

## 5. RECOMMENDATION

Deploy **Linear Regression; persistence is a close simple benchmark** for ultra-short +1h forecasting, leveraging its low computational latency and superior immediate accuracy.